Pharmaceutical Formulation 1

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Formulation 1 (A09662)

Short Title:	Pharmaceutical Formulation 1
Faculty	Science and Computing
Department	Science
Cluster	Pharmaceutical Science
Author	ADONEGAN
Credits	10

Level: Intermediate

Description of Module / Aims

This module will cover the main technologies related to finished product pharmaceutical formulation. Strategies for formulation of solid oral dosage forms and disperse systems will be discussed, and students will carry out an integrated set of experiments thereon.

Programmes

PHAR-0003	BSc in Good Manufacturing Practice and Technology (WD_SGMPT_D)	1 / 2 / M

Indicative Content

- Pharmaceutical dosage forms solid oral dosage forms and disperse systems
- Excipient types and functions: material properties
- Pre-formulation studies
- Formulation development: solid dosage forms; propose and assess simple formulations
- Finished dosage form testing: dissolution profiles
- Stability testing
- Effects of particle size of blending and dissolution
- Suspensions formulation and testing

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Classify the properties and uses of common pharmaceutical excipients.
2. Propose a protocol for a pre-formulation study on an API in advance of formulation trials for a range of dosage forms.
3. Apply the principles of preformulation to simple mixtures to produce formulations.
4. Construct a protocol for the stability testing of a pharmaceutical product.
5. Demonstrate the operation and uses of dissolution apparatus and techniques and complete a dissolution test to determine drug release.
6. Propose a formulation for both a stable suspension and solid oral dosage forms with appropriate choice of excipients.
7. Plan appropriate specifications for the control of the quality of the dosage forms studied.

Learning and Teaching Methods

- Lectures.
- Practicals.
- Industrial visits.
- E-learning via Moodle.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,6,7
Continuous Assessment	50%	
Lab Report	25%	3,5
In-Class Assessment	25%	1,2,4

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Lab		12
Independent Learning		234

Essential Material

- "EU Legislation." ec.europa.eu./health/documents/eudralex. ec.europa.eu./health/documents/eudralex
- "ICH guidelines." https://www.ich.org

Supplementary Material

- Lachman, L. _The theory and Practice of industrial pharmacy_. 3rd ed. Philadelphia: Lea and Febiger, 1986.
- Rowe, R.C. _Handbook of Pharmaceutical excipients_. London: Pharmaceutical Press, 2012.

Requested Resources

- Room Type: Chemistry Lab